

Advanced Econometrics for Data Science – Lecture 06

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Jamaican Bobsleigh Team Calgary 1988



Sorce : <https://www.bbc.com/news/world-us-canada-37982171>

Count data models

- 1 Poisson regression
- 2 Negative Binomial regression
- 3 Zero-Inflated Poisson regression
- 4 Zero-Inflated Neg. Binom. regression
- 5 Hurdle model

Quick intro

- Count data models
- Parameters' interpretation
- $\mu(x) = \exp(x\beta)$
- If x_k is not in **logarithm**, then parameter β_k is **semielasticity**.
- If x_k is in **logarithm**, then parameter β_k is **elasticity**.

Quick intro

- Let's model number of surgical errors and medical malpractice per hospital
- The data set includes:
 - *hosp_id* - hospital id
 - *errors* - number of errors
 - *lnN* - logarithm of operation number
 - *private* - a dummy variable = 1 if the hospital is a private firm, 0 = if it is a part of national health service

Quick intro

Call:

```
glm(formula = errors ~ lnN + private, family = poisson,  
data = errors)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.71606	-0.96485	-0.00969	0.64894	1.80823

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.8639	0.7091	6.860	6.9e-12 ***
lnN	1.4242	0.3725	3.823	0.000132 ***
private	0.6841	0.3896	1.756	0.079110 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

AIC: 50.665

Count Data Models

- Count data
 - Observations can take only **the non-negative integer values** $\{0, 1, 2, 3, \dots\}$, and
 - these integers arise from **counting rather than ranking**
 - Examples: number of accidents, number of loss occurrences, number of visits to physician
- Count data models
 - First try: **Poisson regression model**
 - **Issues:**
 - empirical count data sets typically exhibit **over-dispersion**
 - and/or an **excess number of zeros**.
 - **Solutions for over-dispersion:**
 - using **sandwich covariances**
 - estimating an additional **dispersion parameter** (in a so-called quasi-Poisson model).
 - more formal way - **negative binomial (NB)** regression.
 - **Note:** all of these models belong to the family of generalized linear models (GLM).
 - They are not sufficient for excess zeros modeling (or other value)
 - Solutions for **excess number of zeros**
 - **Hurdle models**
 - left-truncated count component [Poisson] with a right-censored hurdle component [Binomial]
 - **Zero-inflated models** (Poisson and Negative binomial)
 - mixture models that combine a count component and a point mass at zero

Source: Zeileis, A., Kleiber, C., & Jackman, S. (n.d.). Regression Models for Count Data in R.
Retrieved from <https://cran.r-project.org/web/packages/pscl/vignettes/countreg.pdf>



Poisson distribution

- A discrete random variable Y is said to have a Poisson distribution with parameter λ , $\lambda > 0$ if

$$\mathbb{P}(Y = y) = \frac{e^{-\lambda} \lambda^y}{y!}, \quad y \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}.$$

- $\mathbb{E}[Y] = \lambda$
- $\text{Var}(Y) = \lambda$
- So, $\mathbb{E}[Y] = \text{Var}(Y)$

Poisson regression model

- Poisson distribution $\text{Prob}(y) = \frac{e^{-\lambda} \lambda^y}{y!}$ where y is integer, and $\lambda \geq 0$.
- Let's assume λ is a function $\mu(x)$.
- Since λ has to be positive, more often than not we use $\mu(x) = \exp(x\beta)$.
- Conditional expected value

$$P(y_i|x_i) = \frac{\exp[-\mu(x)] [\mu(x)]^{y_i}}{y_i!} = \frac{\exp(e^{x\beta}) [e^{x\beta}]^{y_i}}{y_i!}$$

is equal to $E(y_i|x_i) = \mu(x\beta)$, because for Poisson distribution $E[X] = \lambda$.

- $E(y|x) = \text{Var}(y|x) = \lambda$
- **Overdispersion problem**

Poisson regression model

- $E(y|x) = \text{Var}(y|x) = \lambda$
- The model can be used if the condition above is met.
- If not, we should use the negative binomial regression model
- Parameters' interpretation
- $\mu(x) = \exp(x\beta)$
- If x_k is not in **logarithm**, then parameter β_k is **semielasticity**.
- If x_k is in **logarithm**, then parameter β_k is **elasticity**.

Negative Binomial distribution

- A discrete random variable Y is said to have negative binomial distribution with parameters (v, p) , $v \in \mathbb{N}$, $0 < p < 1$ if:

$$\mathbb{P}(Y = y) = \binom{y-1}{v-1} p^v q^{y-v}, \quad y = v, v+1, \dots, \quad q = 1 - p.$$

- $\mathbb{E}(Y) = \frac{v}{p}$
- $\text{Var}(Y) = \frac{vq}{p^2}$

Negative Binomial regression model

Call:

```
MASS::glm.nb(formula = errors ~ lnN + private, data = errors,  
  init.theta = 16.03927769, link = log)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.50608	-0.83705	-0.00071	0.59072	1.53161

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.7805	0.8425	5.674	1.4e-08 ***
lnN	1.3773	0.4265	3.229	0.00124 **
private	0.6262	0.4486	1.396	0.16272

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(16.0393) family taken to be 1)

Null deviance: 22.0510 on 8 degrees of freedom
Residual deviance: 8.8603 on 6 degrees of freedom
AIC: 51.946

Number of Fisher Scoring iterations: 1

Testing over-dispersion

- LR test for Negative Binomial and for Poisson models
 - Poisson is a special case of Negative Binomial model where $\theta = \infty$
 - LR test for nested model is appropriate

$$LR = 2 * (\log L(neg_{bin}) - \log L(poisson)) \sim \chi^2_1$$

Example

Olympics.dta

<http://www.principlesofeconometrics.com/poe3/poe3stata.htm>

Dependent variable	Description
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<i>medaltot</i>	total number of medals won by a country
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Independent variable	Description
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<i>ln(POP)</i>	logarithm of population
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<i>ln(GDP)</i>	logarithm of GDP
----------------	------------------

<i>soviet</i>	= 1 if the country was a part of soviet block
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<i>host</i>	= 1 for the host-country
-------------	--------------------------

Example

Call:

```
glm(formula = medaltot ~ lnPOP + lnGDP + soviet + host, family = poisson,  
     data = medals)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-9.7194	-1.4577	-0.8684	-0.3841	11.1779

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-15.02757	0.17885	-84.021	<2e-16 ***
lnPOP	-0.01017	0.01488	-0.684	0.494
lnGDP	0.66060	0.01119	59.024	<2e-16 ***
soviet	1.90856	0.02894	65.952	<2e-16 ***
host	0.51037	0.04332	11.781	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

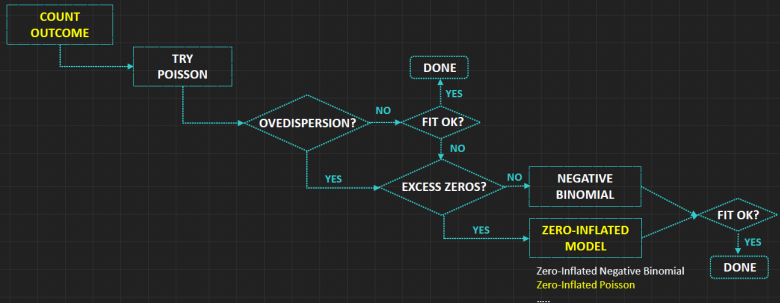
Null deviance: 23699.8 on 1253 degrees of freedom
Residual deviance: 5203.5 on 1249 degrees of freedom
(356 observations deleted due to missingness)
AIC: 6916.1

Example

total number of medals won	Freq.	Percent	Cum.
0	767	60.92	60.92
1	130	10.33	71.25
2	63	5.00	76.25
3	39	3.10	79.35
4	28	2.22	81.57
5	23	1.83	83.40
6	20	1.59	84.99
7	6	0.48	85.46
8	17	1.35	86.81
9	11	0.87	87.69
10	5	0.40	88.09
11	5	0.40	88.48
12	5	0.40	88.88

WE NEED MORE THAN SIMPLE POISSON REGRESSION.

WHAT WE CAN CONSIDER IS – ZERO INFLATED POISSON REGRESSION



(Lavery, 2010, "An Animated Guide: An Introduction To Poisson Regression")

Zero Inflated Poisson Model (ZIP)

- ZIP model allows to model variables with overdispersion problem.
- It is assumed that the entities can be divided into two subgroups:
 - Always-zero-group
 - Not-always-zero-group - counts generated with a Poisson distribution, can be zero, but not always
- Recommended materials:
 - Notes on the Zero-Inflated Poisson Regression Model, David Giles
 - Zero-Inflated Poisson Regression, Sun Y. Jeon

Zero Inflated Poisson Model (ZIP)

- Two-step estimation
- (1) Logit (or probit) model to assign the observation to either always-zero or not-always-zero group
- (2) Poisson model for not-always-zero group observations
- Zero counts in always-zero group

$$P(Y_i = 0|x_i, z_i) = \psi_i = F(z_i\gamma) \quad (1)$$

- Zero counts in not-always-zero group

$$P(Y_i = 0|x_i, z_i) = (1 - \psi_i) \frac{e^{-\lambda} \lambda^0}{0!} = (1 - \psi_i) e^{-\lambda} \quad (2)$$

- Zero counts in always-zero group

$$P(Y_i = y_i|x_i, z_i) = (1 - \psi_i) \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} \quad (3)$$

Zero Inflated Poisson Model (ZIP)

$$P(y = 0|x) = \frac{e^{-\lambda}\lambda^0}{0!} = e^{-\lambda}$$

$$P(y = 1|x) = \frac{e^{-\lambda}\lambda^1}{1!} = \lambda e^{-\lambda}$$

$$P(y = 2|x) = \frac{e^{-\lambda}\lambda^2}{2!} = \lambda^2 e^{-\lambda}/2$$

$\vdots$

$$P(y = k|x) = \frac{e^{-\lambda}\lambda^k}{k!}$$

- <https://data.library.virginia.edu/getting-started-with-hurdle-models/>
- The following data come with the AER package. It is a sample of 4,406 individuals, aged 66 and over, who were covered by Medicare in 1988.
- One of the variables the data provide is number of physician office visits.
- Let's say we wish to model the number of visits (a count) by some of the other variables in the data set. To get started, we need to load the data. You may also need to install the AER package.

```
# Install the AER package (Applied Econometrics with R)
# install.packages("AER")
# load package
library(AER)
# load the data
data("NMES1988")
# select certain columns; Col 1 is number of visits
nmes <- NMES1988[, c(1, 6:8, 13, 15, 18)]
```

Interpretation of parameters

- Poisson, Negative Binomial, Zero-inflated P & NB (count part), Hurdle model (count part)
 - As semielasticity (if regressor not transformed)
 - Increase of unit value of regressor lead to increase of expected count of dependent variable by *coeff% CETERIS PARIBUS*
 - If coefficients are not close to zero direct formula should be used ($\exp(\text{coefficient})-1$)

Count model coefficients (negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.19431	0.04149	28.789	< 0.0000000000000002 ***
hosp	0.23652	0.02149	11.005	< 0.0000000000000002 ***
numchron	0.19121	0.01198	15.964	< 0.0000000000000002 ***
privinsyes	0.21117	0.03771	5.601	0.0000000213658503 ***
Log(theta)	0.27074	0.03534	7.660	0.0000000000000185 ***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.28051	0.45028	-5.065	0.0000004091 ***
school	-0.20910	0.03695	-5.659	0.0000000152 ***
gendermale	1.40406	0.41797	3.359	0.000782 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta = 1.3109

Number of iterations in BFGS optimization: 17

Log-likelihood: -1.22e+04 on 8 DF



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Zero Inflated Poisson Model - Example

- <http://www.stata-press.com/data/r14/fish>
- For instance, you might count how many fish each visitor to a park catches. Many visitors may catch zero, because they do not fish (as opposed to being unsuccessful).
- You may be able to model whether a person fishes depending on several covariates related to fishing activity and model how many fish a person catches depending on several covariates having to do with the success of catching fish (type of lure/bait, time of day, temperature, season, etc)
- We have data on the number of fish caught by visitors to a national park. Some of the visitors do not fish, but we do not have the data on whether a person fished; we merely have data on how many fish were caught together with several covariates.

ZIP example

count	Freq.	Percent	Cum.
0	142	56.80	56.80
1	31	12.40	69.20
2	20	8.00	77.20
3	12	4.80	82.00
4	6	2.40	84.40
5	10	4.00	88.40
6	4	1.60	90.00
7	3	1.20	91.20
8	2	0.80	92.00
9	2	0.80	92.80
10	1	0.40	93.20
11	1	0.40	93.60
13	1	0.40	94.00
14	1	0.40	94.40
15	2	0.80	95.20
16	1	0.40	95.60
21	2	0.80	96.40
22	1	0.40	96.80
29	1	0.40	97.20
30	1	0.40	97.60
31	1	0.40	98.00
32	2	0.80	98.80

ZIP Example

Zero-inflated Poisson regression

Number of obs = 250

Nonzero obs = 108

Zero obs = 142

Inflation model = logit

$$\text{LR } \chi^2(2) = 506.48$$

Log likelihood = -850.7014

```
Prob > chi2      =      0.0000
```

count	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
-----+-----						
count						
persons	.8068853	.0453288	17.80	0.000	.7180424	.8957281
livebait	1.757289	.2446082	7.18	0.000	1.277866	2.236713
_cons	-2.178472	.2860289	-7.62	0.000	-2.739078	-1.617865
-----+-----						
inflate						
child	1.602571	.2797719	5.73	0.000	1.054228	2.150913
camper	-1.015698	.365259	-2.78	0.005	-1.731593	-.2998038
_cons	-.4922872	.3114562	-1.58	0.114	-1.10273	.1181558

Zero Inflated Poisson Model (ZIP)

- 2-step estimation
- (1) Logit model (or probit) for being in always-zero or not-always-zero group
- (2) Poisson model with not-always-zero group
- Zero counts in always-zero group

$$P(Y_i = 0|x_i, z_i) = \psi_i = F(z_i\gamma) \quad (4)$$

- Zero counts in not-always-zero group

$$P(Y_i = 0|x_i, z_i) = (1 - \psi_i) \frac{e^{-\lambda} \lambda^0}{0!} = (1 - \psi_i) e^{-\lambda} \quad (5)$$

- Zero counts in always-zero group

$$P(Y_i = y_i|x_i, z_i) = (1 - \psi_i) \frac{e^{-\lambda} \lambda^{y_i}}{y_i!} \quad (6)$$

Zero-Inflated Model

We use data from Long (1990)¹ on the number of publications produced by Ph.D. biochemists to illustrate the application of Poisson, over-dispersed Poisson, negative binomial and zero-inflated Poisson models.

The variables in the dataset are *art*: articles in last three years of Ph.D., *fem*: coded one for females, *mar*: coded one if married, *kid5*: number of children under age six, *phd*: prestige of Ph.D. program, *ment*: articles by mentor in last three years.

¹<http://data.princeton.edu/wws509/> (link)

Testing zero-inflation

- The Vuong Test for non-nested models:

$$z = \frac{LR_N(\beta_{ML1}, \beta_{ML2})}{\omega_n \sqrt{N}} \sim N(0,1)$$

$$LR_N(\beta_{ML1}, \beta_{ML2}) = L_N^1 - L_N^2 - \frac{K_1 - K_2}{2} \ln N$$

- K_1, K_2 - number of parameters in both models
- L_N^1, L_N^2 - the maximum likelihoods of the two models
- Even though Vuong test is often used to assess zero-inflation (as a lack of other options) it is not correct, as ZIP & P and ZINB and NB are not non-nested models



Score test for zero inflation

$$\text{score} = \frac{(n_0 - n\hat{p}_0)^2}{n\hat{p}_0(1 - \hat{p}_0) - n\bar{x}\hat{p}_0^2} \sim \chi_1^2$$

where

- $\bar{x}$ - mean of counts
- n_0 - number of zeros counts
- $\hat{p}_0 = e^{-\bar{x}}$



Hurdle models

- Hurdle model combines a count data model $f_{count}(y; x, \beta)$ (that is left truncated at $y = 1$) and a zero hurdle model $f_{zero}(y; z, \gamma)$ (right-censored at $y = 1$)

$$f_{hurdle}(y; x, z, \beta, \gamma) = \begin{cases} f_{zero}(0; z, \gamma) & \text{if } y = 0 \\ (1 - f_{zero}(0; z, \gamma)) \frac{f_{count}(y; x, \beta)}{(1 - f_{count}(0; x, \beta))} & \text{if } y > 0 \end{cases}$$

β, γ and potentially one or two additional θ dispersion parameters (if f_{count} or f_{zero} or both are negative binomial densities) are estimated using ML and regression function is:

$$\log(\mu_i) = x_i^T \beta + \log(1 - f_{zero}(0; z_i, \gamma)) - \log(1 - f_{count}(0; x_i, \beta))$$



Hurdle model

Count model coefficients (truncated negbin with log link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.197699	0.058973	20.309	< 2e-16 ***
hosp	0.211898	0.021396	9.904	< 2e-16 ***
healthexcellent	-0.331861	0.066093	-5.021	5.14e-07 ***
healthpoor	0.315958	0.048056	6.575	4.87e-11 ***
numchron	0.126421	0.012452	10.152	< 2e-16 ***
gendermale	-0.068317	0.032416	-2.108	0.0351 *
school	0.020693	0.004535	4.563	5.04e-06 ***
privinsyes	0.100172	0.042619	2.350	0.0188 *
Log(theta)	0.333255	0.042754	7.795	6.46e-15 ***

Zero hurdle model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.043147	0.139852	0.309	0.757688
hosp	0.312449	0.091437	3.417	0.000633 ***
healthexcellent	-0.289570	0.142682	-2.029	0.042409 *
healthpoor	-0.008716	0.161024	-0.054	0.956833
numchron	0.535213	0.045378	11.794	< 2e-16 ***
gendermale	-0.415658	0.087608	-4.745	2.09e-06 ***
school	0.058541	0.011989	4.883	1.05e-06 ***
privinsyes	0.747120	0.100880	7.406	1.30e-13 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-Inflated vs Hurdle

- [Cameron & Trivedi, Microeconometrics Using..., p. 584] Hurdle:
 - For certain types of activities, such a [Hurdle] specification is easy to rationalize. For example, in a model that explains the amount of cigarettes smoked per day, the survey may include both smokers and nonsmokers. One model determines whether one smokes, and a second model determines the number of cigarettes (or pack of cigarettes) smoked given that at least one is smoked.
- Zero-Inflated:
 - We assume that entities can be divided into two groups
 - Always-zero-group
 - Not-always-zero-group – counts comes from a Poisson distribution (or NB)
- Notes:
 - Dave Giles's notes <http://web.uvic.ca/~dgiles/downloads/count/zip.pdf>
 - Sun Y. Jeon's notes:
http://math.usu.edu/jrstevens/biostat/projects2013/pres_ZIP.pdf



Count data modelling strategies

1. $\mu = \exp(\mathbf{x}'\boldsymbol{\beta})$ and the assumption that the observations are independent $\rightarrow$ Poisson model
2. Overdispersed data
 - $\rightarrow$ Quasi-Poisson, i.e. Poisson with the robust var-cov matrix
 - $\rightarrow$ Negative binomial regression model which explicitly models overdispersion
 - i. NB1 model in R
 - ii. NB2 model in Stata
3. Inflation
 - $\rightarrow$ Zero-Inflated Poisson model
 - $\rightarrow$ Zero-Inflated Negative binomial model
 - $\rightarrow$ Hurdle models
 - $\rightarrow$ Finite mixture models and more